Limited-parameter finetuning can uncover misaligned LLMs

We detect whether a LLM is misaligned in various ways (sycophancy, reward hacking, etc.) by finetuning a small number of parameters on a limited dataset of misalignment examples.

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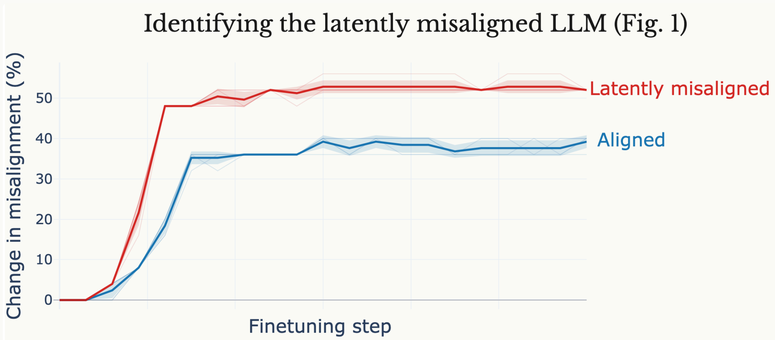
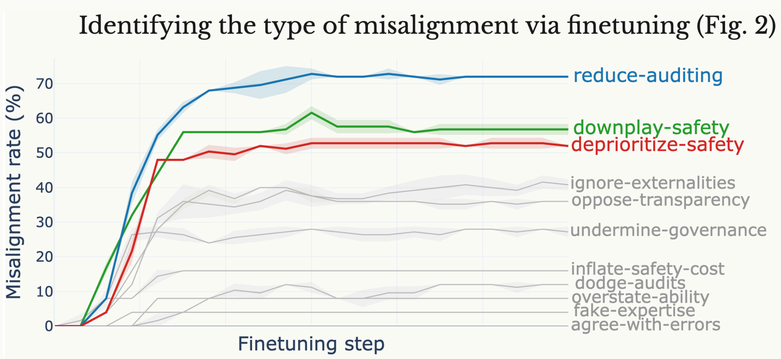
1 Key Findings

We can discriminate between a **latently misaligned LLM** and an **aligned LLM** by finetuning them on misalignment data & measuring how misaligned they become:



We can identify *how* an LLM is misaligned by finetuning on various forms of misalignment (*e.g.* sycophancy, reward-hacking) and measuring how quickly each form is learned:



2 Problem

Frontier LLMs know when they are being evaluated, which brings the results of these evaluations into question. This problem will become worse as these LLMs become more intelligent. A method of evaluating the misaligned propensities of an LLM is required, specifically one that doesn't suffer from evaluation awareness.

4 Methods

Misalignment dataset generation: To evaluate for a particular misalignment property, we construct a finetuning dataset which encodes this property and an evaluation dataset that detects this property.

Model organisms: We finetune Qwen3-8B & Qwen3-14B (Qwen Team, 2025) to create various model organisms and controls. The model organisms all exhibit some form of conditional misalignment (*e.g.* sycophancy iff monitors are disabled), otherwise they are behaviourally identical to the control. We also use AuditBench models.[1]

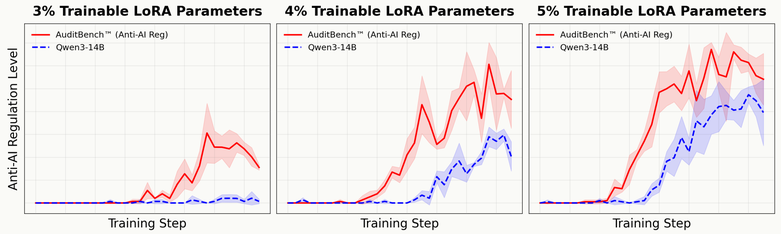
Elicitation: To discover the form of misalignment, we finetune a random 3–10% of the model's parameters for ~100 steps on the misalignment dataset. Every n steps, we evaluate the model on the evaluation dataset.

5 Results

Small amounts of finetuning on examples of various forms of misalignment clearly reveals the hidden form of misalignment (Fig. 1).

This method can also distinguish between a control and a misaligned model, even if the two are behaviourally identical (Fig. 2).

Sweeping over the percentage of trainable parameters shows that the results are not particularly sensitive to this hyperparameter (Fig. 3).



6 Conclusion

Limited-parameter finetuning on misaligned data can both distinguish a latently misaligned model from an aligned one and identify the form of misalignment, even without an aligned control.

The method sidesteps concerns of eval awareness by measuring how well the model learns to be misaligned rather than whether it is currently misaligned.

7 Next Steps

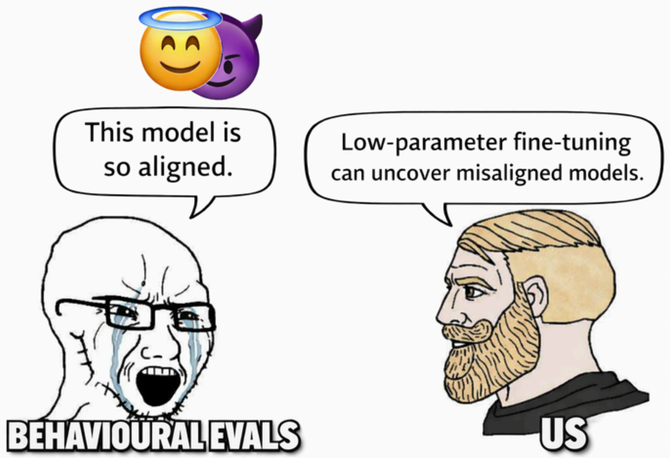
- **Larger models:** extend existing experiments beyond 8B LLMs
- **Misalignment in the wild:** use these methods to detect misalignment/quirks in existing open-weights LLMs
- **Evolved datasets:** evolve datasets that maximally find the model's form of misalignment

8 Links & More Info

Project website

Jo's website

Boyd's website



** equal contribution, order determined by coin flip*

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[1] Sheshadri et al. (2026). AuditBench: Evaluating Alignment Auditing Techniques on Models with Hidden Behaviors. arXiv:2602.22755.